

Advanced Probabilistic Machine Learning

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Graphical Models

Overview

1. Graphical Models
2. Directed Graphical Models: Bayesian Networks
3. Undirected Graphical Models: Markov Random Fields

1. **Graphical Models**
2. Directed Graphical Models: Bayesian Networks
3. Undirected Graphical Models: Markov Random Fields

Graphical Models

■ **Challenge:** How to formulate complex likelihoods/data models & priors for *actual* data?

□ **Example 1:** Match outcomes $y \in \{-1, 1\}$ (data) for a head-to-head match between two players

- **Prior:** $p(\mathbf{s}) = \mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2)$ ← skill belief
 - **Likelihood:** $p(y|\mathbf{s}) = \int \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0) dp_1 dp_2$ ← marginalization
- Match outcome
Player performance

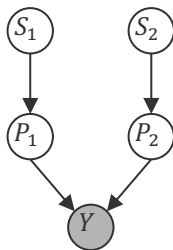
□ **Example 2:** Time series $\mathbf{y}$ of temperatures

- **Prior:** $p(w) = \mathcal{N}(w; \mu, \sigma^2)$ ← External state mapping parameter belief
 - **Likelihood:** $p(\mathbf{y}|w, X) = \int \mathcal{N}(z_1; w \cdot x_1, \tau^2) \cdot \mathcal{N}(y_1; z_1, \beta^2) \cdot \mathcal{N}(z_2; z_1 + w \cdot x_2, \tau^2) \cdots dz$ ← marginalization
- Advanced Probabilistic Machine Learning
- Unit 3 – Graphical Models
- Dynamics model
Observed temperature model
Conditional hidden state model

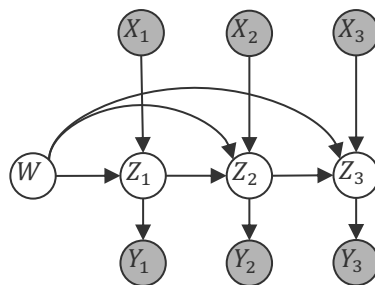
Graphical Models

- **Observation:** The product structure of the probabilities seems crucial
- **Idea:** Define a graph where each of the variables are nodes and edges indicate factor relationships between variables

$$\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)$$



$$\mathcal{N}(w; \mu, \sigma^2) \cdot \mathcal{N}(z_1; w \cdot x_1, \tau^2) \cdot \mathcal{N}(y_1; z_1, \beta^2) \cdot \mathcal{N}(z_2; z_1 + w \cdot x_2, \tau^2) \cdot \mathcal{N}(y_2; z_2, \beta^2) \dots$$



- **Advantages:** Simple way to visualize factor structure of the joint probability
 - **Graphical Models:** Conditional independence based on graph properties
 - **Factor Graphs:** Efficient inference and approximation algorithms (next week)

Conditional Independence

- In modelling specific data, domain experts often know whether two (latent) measurements can affect each other or not (i.e., are independent)
 - **Examples:**
 - Skills of two players in a video game are not dependent if they never played before
 - Skills of two players in a video game *are* dependent if they have played many times!



Philip Dawid
(1946–)

- Graphical models are useful to determine conditional independence.
- **Conditional Independence.** A random variable X_i is conditionally independent of a random variable X_j given the variable X_k if for all values X_k

$$p(X_i = x_i | X_j = x_j, X_k = x_k) = p(X_i = x_i | X_k = x_k)$$

- Equivalent definition: $p(X_i = x_i, X_j = x_j | X_k = x_k) = p(X_i = x_i | X_k = x_k) \cdot p(X_j = x_j | X_k = x_k)$
- Shorthand notation (Dawid, 1979): $X_i \perp X_j | X_k$

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Unit 3 – Graphical Models

Overview

1. Graphical Models
- 2. Directed Graphical Models: Bayesian Networks**
3. Undirected Graphical Models: Markov Random Fields

Bayesian Networks

- **Observation.** Any joint distribution $p(x_1, \dots, x_n)$ can be written as

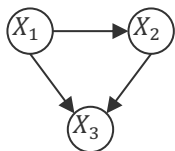
$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | x_1, \dots, x_{i-1})$$

- **Bayesian Network.** Given a joint distribution

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | \mathbf{x}_{\text{parents}_i})$$

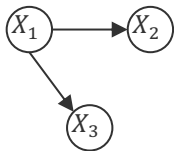
a Bayesian network is a graph with a node for every variable X_i , and a directed edge from every variable $X_j, j \in \text{parent}_i$ to X_i .

- **Examples:** For 3 variables, we have these four generic Bayesian networks



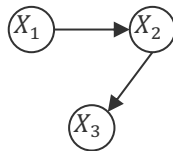
$$p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2 | x_1) \cdot p(x_3 | x_1, x_2)$$

full mesh



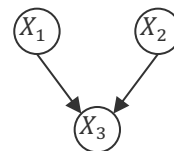
$$p(x_1, x_2, x_3) = p(x_1) \cdot \prod_{i=2}^3 p(x_i | x_1)$$

star



$$p(x_1, x_2, x_3) = p(x_1) \cdot \prod_{i=2}^3 p(x_i | x_{i-1})$$

chain



$$p(x_1, x_2, x_3) = p(x_3 | x_1, x_2) \cdot \prod_{i=1}^2 p(x_i)$$

sink

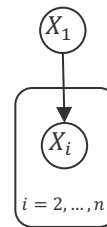
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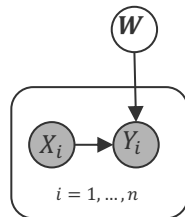
Bayesian Network Models

- **Plate.** If a subset of variables has the same relation only differing in their index, we use a "plate" to collapse them into a single graphical element.
 - Increase readability of models for large amounts of parameters and data
- A Bayesian network must always be a **directed acyclic graph** because only those have a topological order corresponding to a variable order.
- **Observed Variables.** If a subset of variables has been observed ("data"), the variable nodes are usually shaded ("clamped").
 - **Example:** Discriminatory Models

$$p(x_1, x_2, \dots, x_n) = p(x_1) \cdot \prod_{i=2}^n p(x_i | x_1)$$



$$p(\mathbf{w}, y_1, \dots, y_n | x_1, \dots, x_n) = \prod_{i=1}^n p(y_i | x_i, \mathbf{w}) \cdot p(\mathbf{w})$$



$$p(\mathbf{w} | (x_1, y_1), \dots, (x_n, y_n)) \propto \prod_{i=1}^n p(y_i | x_i, \mathbf{w}) \cdot p(\mathbf{w})$$

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Representation Complexity

- For simplicity, let us assume that $x_i \in \{1, \dots, K\}$

Naive

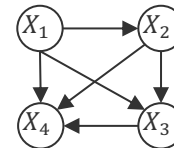
$$p(x_1, \dots, x_n)$$

x_1	x_2	x_3	x_4	$p(x_1, x_2, x_3, x_4)$
1	1	1	1	p_{1111}
1	1	1	2	p_{1112}
$\vdots$				
1	1	1	K	p_{111K}
1	1	2	1	p_{1121}
$\vdots$				
K	K	K	K	$1 - \Sigma$

$$K^4 - 1$$

Bayesian Network

$$p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1, x_2) \cdot p(x_4|x_1, x_2, x_3)$$



x_1	$p(x_1)$
1	p_1
2	p_2
$\vdots$	
K	$1 - \Sigma$

$$K - 1$$

x_2	x_1	$p(x_2 x_1)$
1	1	p_{11}
2	1	p_{21}
$\vdots$		
K	1	$1 - \Sigma$
1	2	p_{12}
$\vdots$		
K	K	$1 - \Sigma$

$$(K - 1) \cdot K$$

x_3	x_1	x_2	$p(x_3 x_1, x_2)$
1	1	1	p_{111}
2	1	1	p_{211}
$\vdots$			
K	1	1	$1 - \Sigma$
1	1	2	p_{112}
$\vdots$			
K	K	K	$1 - \Sigma$

$$(K - 1) \cdot K^2$$

x_4	x_1	x_2	x_3	$p(x_4 x_1, x_2, x_3)$
1	1	1	1	p_{1111}
2	1	1	1	p_{2111}
$\vdots$				
K	1	1	1	$1 - \Sigma$
1	1	1	2	p_{1112}
$\vdots$				
K	K	K	K	$1 - \Sigma$

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$$(K - 1) \cdot K^3$$

$$\begin{aligned}
 (K - 1) \cdot (1 + K + K^2 + K^3) &= (K + K^2 + K^3 + K^4) - (1 + K + K^2 + K^3) \\
 &= K^4 - 1
 \end{aligned}$$

Sampling a Bayesian Network

- One advantage of a Bayesian network is the ability to *sample* $p(x_1, \dots, x_n)$

Ancestral Sampling

1. Topologically sort all variables $X_1, \dots, X_n$ into $X_{(1)}, \dots, X_{(n)}$
2. Sample each variable $X_{(i)}$ using distribution $p(X_{(i)} | X_{(1)}, \dots, X_{(i-1)})$

- **Assumption**

1. Sampling from the conditional distributions is simpler than from the joint distribution
2. There are no clamped nodes, that is, we do not condition on any variable

- **Problems**

1. Sampling is *sequential* one variable at the time
2. Conditioning happens only on frequent events because for samples $x_{j,1}, \dots, x_{j,n}$

$$\frac{|\{(x_{j,1}, \dots, x_{j,n}) \mid x_{j,1} = x_1 \wedge \dots \wedge x_{j,n} = x_n\}|}{|\{x_{j,n} = x_n\}|} \approx \frac{p(x_1, \dots, x_n)}{p(x_n)} = p(x_1, \dots, x_{n-1} | x_n)$$

≈ 0 for rare events

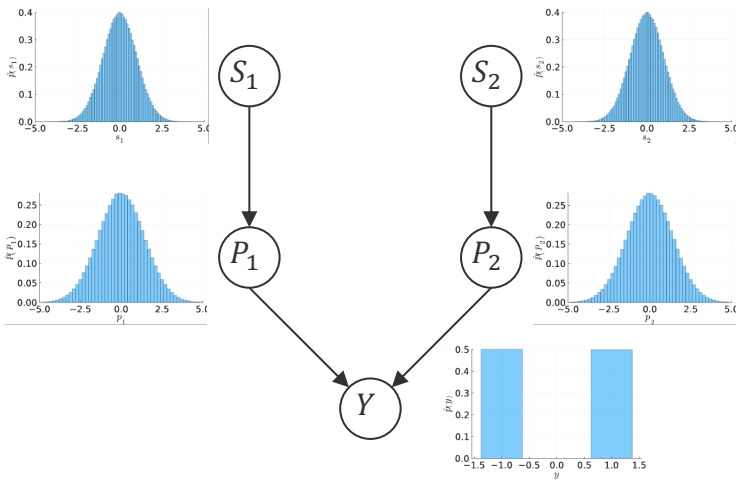
Sampling a Bayesian Network: Example

$$\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)$$

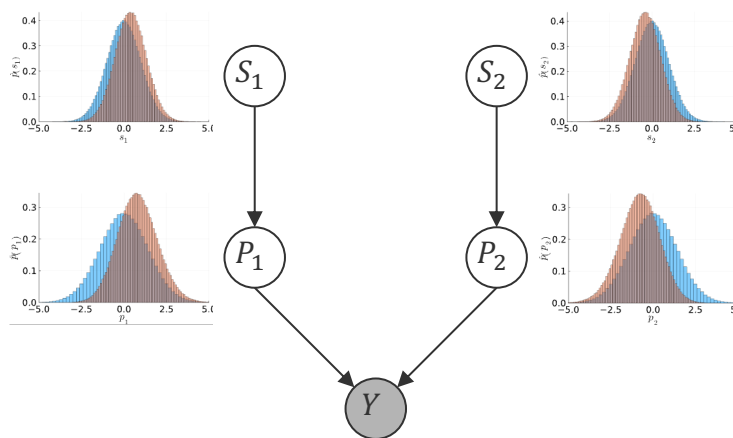
$$\mu_1 = \mu_2$$

```
# samples from the TrueSkill graphical model
function sample(; n = 100000, μ1=0.0, σ1=1.0, μ2=0.0, σ2=1.0, β=1.0)
    samples = Vector{Vector{Float64}}(undef, n)
    for i in 1:n
        s1 = rand(Normal(μ1, σ1))
        s2 = rand(Normal(μ2, σ2))
        p1 = rand(Normal(s1, β))
        p2 = rand(Normal(s2, β))
        y = p1 > p2 ? 1.0 : -1.0
        samples[i] = [s1, s2, p1, p2, y]
    end
    return samples
end
```

Without match outcome



With match outcome ($y = 1$)



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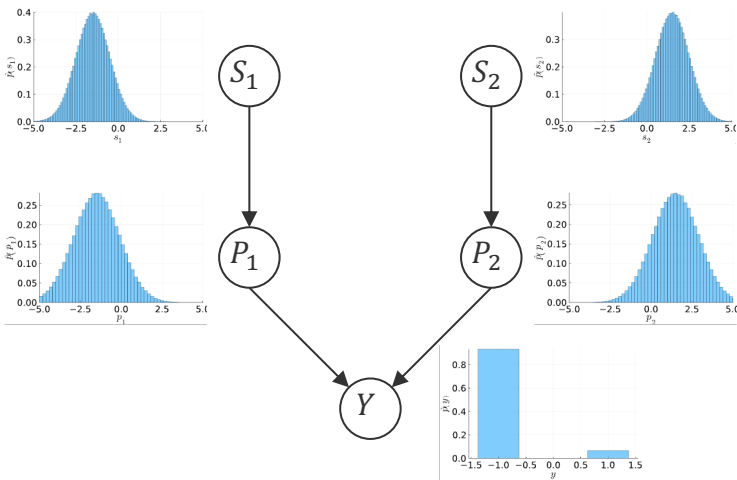
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Sampling a Bayesian Network: Example (ctd)

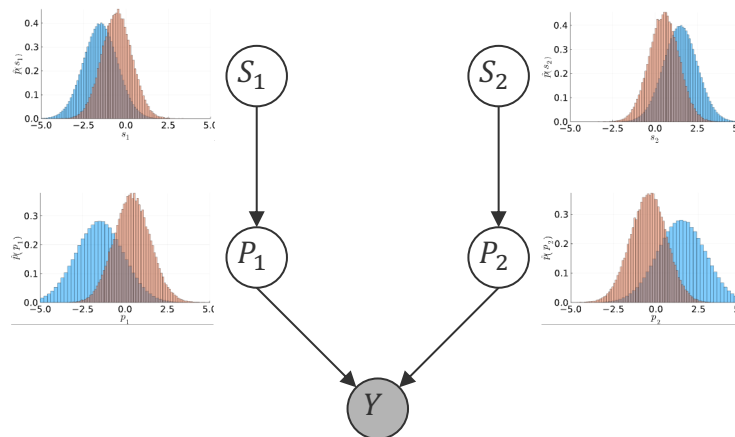
$$\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)$$

$$\mu_1 \ll \mu_2$$

Without match outcome



With match outcome ($y = 1$)

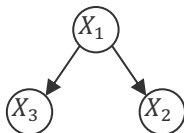


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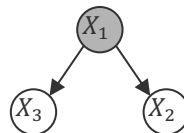
Conditional Independence: Warm-Up I

- **Tail-to-Tail Node (x_1):** $p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1)$



$$p(x_2, x_3) = \sum_{\{x_1\}} p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1) \neq p(x_2) \cdot p(x_3)$$

not independent



$$p(x_2, x_3|x_1) = \frac{p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1)}{p(x_1)} = p(x_2|x_1) \cdot p(x_3|x_1)$$

conditionally independent

- **Head-to-Tail Node (x_2):** $p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_2)$



$$p(x_1, x_3) = p(x_1) \cdot \sum_{\{x_2\}} p(x_2|x_1) \cdot p(x_3|x_2) = p(x_1) \cdot p(x_3|x_1) \neq p(x_1) \cdot p(x_3)$$

not independent



$$p(x_1, x_3|x_2) = \frac{p(x_2|x_1) \cdot p(x_1)}{p(x_2)} \cdot p(x_3|x_2) = p(x_1|x_2) \cdot p(x_3|x_2)$$

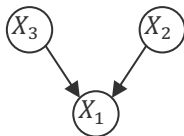
conditionally independent

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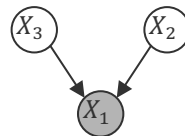
Conditional Independence: Warm-Up II

- **Head-to-Head Node (x_1):** $p(x_1, x_2, x_3) = p(x_2) \cdot p(x_3) \cdot p(x_1|x_2, x_3)$



$$p(x_2, x_3) = \sum_{\{x_1\}} p(x_1|x_2, x_3) \cdot p(x_2) \cdot p(x_3) = p(x_2) \cdot p(x_3)$$

independent

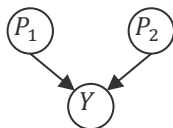


$$p(x_2, x_3|x_1) = \frac{p(x_1|x_2, x_3) \cdot p(x_2) \cdot p(x_3)}{p(x_1)} \neq p(x_2|x_1) \cdot p(x_3|x_1)$$

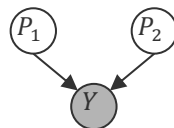
not (always) conditionally independent

- It can be shown that the path between X_2 and X_3 are only independent if *none* of the *descendant* node from X_1 (that can be reached in the directed graph) is observed!

- **Skill Example (ctd):** Consider the performance of two players



Before match: P_1 and P_2 are independent



After match: P_1 and P_2 are **not** independent

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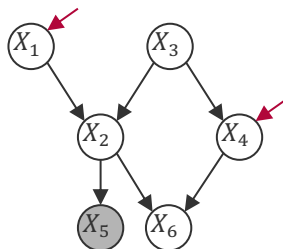
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Conditional Independence: d-separation

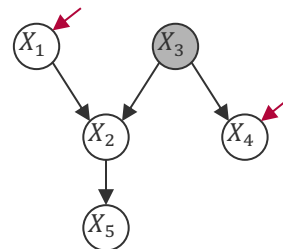
- **Blocked Node.** A node in a Bayesian network is said to be blocked if
 1. It's a head-to-tail or tail-to-tail node and the node is observed.
 2. It's a head-to-head node and neither the node nor any of its descendants are observed.
- **d-separation.** Given a Bayesian network and a subset of observed variables, two non-observed variables X_i and X_j are conditionally independent (that is, d-separated) if every undirected path between X_i and X_j contains at least one blocked node.
- **Examples.**



Judea Pearl
(1936–)



X_1 and X_4 are not independent



X_1 and X_4 are independent

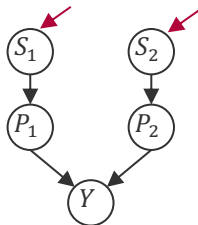
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Conditional Independence: Skill Example

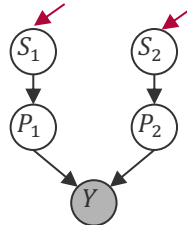
- **Skill Example (ctd):** Consider the skills of two players

Before match



S_1 and S_2 are independent

After match



S_1 and S_2 are **not** independent

- Intuitive because

- Before the match there is no information that “links” the skill of two players
- After the match, if the skill of the winning player goes down (e.g., due to a loss in a subsequent match) then the skill of the opponent also needs to go down (or otherwise the observed match outcome would not have been possible)

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Overview

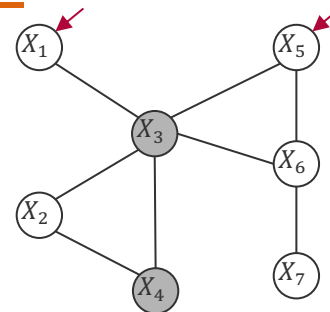
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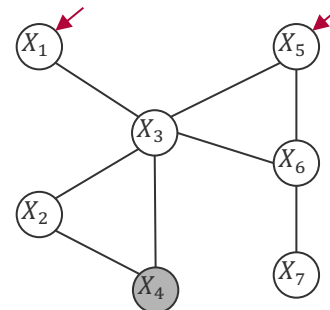
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Conditional Independence Revisited

- **Observation:** Bayesian networks are visual representations of conditional distributions, but conditional independence requires checking a graph property (d-separation) that is
 - based on removing the direction of the arrows
 - not entirely local
- **Idea:** Remove the direction of arrows in a graphical model and define conditional independence by pure connectivity!
- **Conditional Independence** . *Given an undirected graphical model and a subset D of observed variables, two variables X_i and X_j are conditionally independent if there is no path between X_i and X_j when removing all variables $X_k \in D$.*
 - This property is also called the *Markov property* of the graph of random variables X_i
 - **Pro:** Very easy to verify conditional independence of variable sets
 - **Con:** Given this conditional independence definition, what is the functional form of the joint probability that satisfies this condition?



X_1 and X_5 are conditionally independent



X_1 and X_5 are **not** conditionally independent

Hammersley-Clifford Theorem

- **Hammersley-Clifford (1971).** Given n random variables $X_1, \dots, X_n$, a density p satisfies the Markov property for an undirected graph G over $X_1, \dots, X_n$ if and only

$$p(\mathbf{x}) \propto \prod_{C \in \text{cliques}(G)} \psi_C(\mathbf{x}_C)$$

where $\psi_C(\cdot)$ are strictly positive functions and $\mathbf{x}_C$ are the values of $\mathbf{x}$ indexed by C .

- Computing the density up to normalization is computationally easy but the normalization constant is computationally hard as it requires sums/integrals of products of potentials!



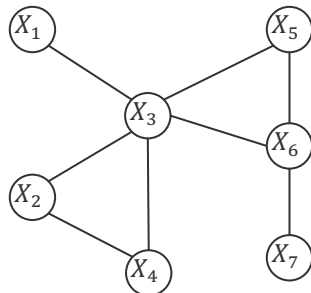
John Hammersley
(1920 – 2004)



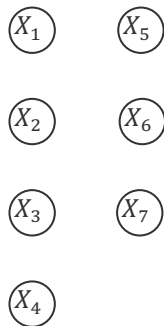
Peter Clifford
(1943 – 2025)

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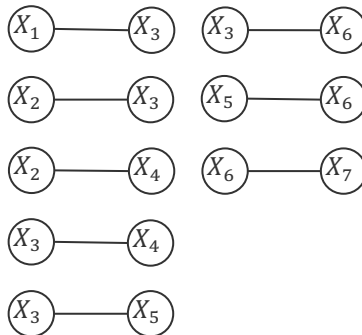
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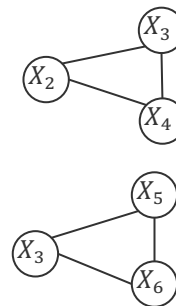
Cliques of size 1



Cliques of size 2



Cliques of size 3



Proof of the Hammersley-Clifford Theorem

- Assume that $p(\mathbf{x}) \propto \prod_{C \in \text{cliques}(G)} \psi_C(\mathbf{x}_C)$ and pick X_i, X_j and D such that removing all variables in D from the graph G separates X_i, X_j . Define $S = V \setminus (D \cup \{X_i\} \cup \{X_j\})$

- Then $\text{cliques}(G)$ separates in three disjoint sets

$$\mathcal{C}_{X_i} := \{C: C \cap \{X_i\} \neq \emptyset \wedge C \cap \{X_j\} = \emptyset\}$$

$$\mathcal{C}_{X_j} := \{C: C \cap \{X_j\} \neq \emptyset \wedge C \cap \{X_i\} = \emptyset\}$$

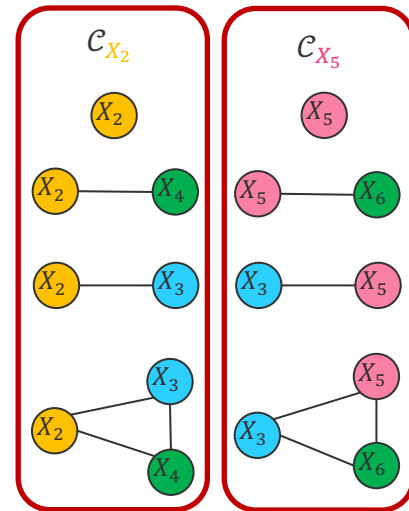
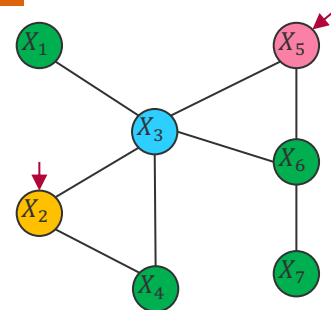
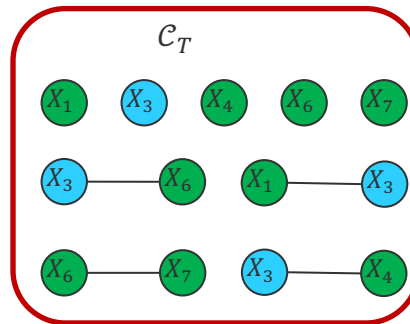
$$\mathcal{C}_T := \{C: C \cap \{X_i\} = \emptyset \wedge C \cap \{X_j\} = \emptyset\}$$

- Thus, the joint density has the following property

$$p(\mathbf{x}) \propto \left[\prod_{C \in \mathcal{C}_{X_i}} \psi_C(\mathbf{x}_C) \right] \cdot \left[\prod_{C \in \mathcal{C}_{X_j}} \psi_C(\mathbf{x}_C) \right] \cdot \left[\prod_{C \in \mathcal{C}_T} \psi_C(\mathbf{x}_C) \right]$$

- Because all variables in D separate X_i, X_j we have $S = S_i \cup S_j \cup S_0$ which are disjoint where S_i are the variables in S which are in cliques in $\mathcal{C}_{X_i}$ (and similar for S_j). Thus

$$p(x_i, x_j | \mathbf{x}_D) \propto \underbrace{\left[\sum_{\{x_{S_i}\}} \prod_{C \in \mathcal{C}_{X_i} \cup \mathcal{C}_T} \psi_C(\mathbf{x}_C) \right]}_{q_i(x_i)} \cdot \underbrace{\left[\sum_{\{x_{S_j}\}} \prod_{C \in \mathcal{C}_{X_j} \cup \mathcal{C}_T} \psi_C(\mathbf{x}_C) \right]}_{q_j(x_j)} \cdot \underbrace{\left[\sum_{\{x_{S_0}\}} \prod_{C \in \mathcal{C}_T} \psi_C(\mathbf{x}_C) \right]}_{\text{constant}}$$



Sampling an Undirected Graphical Model

- If the X_i are discrete random variables, we can *sample* $p(x_1, \dots, x_n)$ as follows

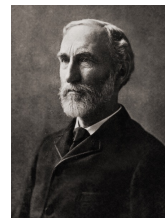
Gibbs Sampling

1. Initialize all random variables with a value $x_1, \dots, x_n$
2. For all $i \in \{1, \dots, n\}$, $x_i \sim p(X_i | X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1}, \dots, X_n = x_n)$
3. Skip K such generated samples $\mathbf{x}$ to remove

- **Assumption:** There are only a few cliques $\mathcal{C}_{x_i} := \{C: C \cap \{X_i\} \neq \emptyset\}$ because

$$p(X_i | X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1}, \dots, X_n = x_n) \propto \prod_{C \in \mathcal{C}_{x_i}} \psi_C(\mathbf{x}_C)$$

- **Problem:** Step 2 produced highly correlated samples (thus we skip K samples $\mathbf{x}$)
- **Advantage:** We can easily condition on observed variables thereby reducing the number of elements of $\mathbf{x}$ in step 2!



Josiah W. Gibbs
(1839 – 1903)



Donald Geman
(1943 –)

Advanced Probabilistic
Machine Learning

Unit 3 – Graphical Models

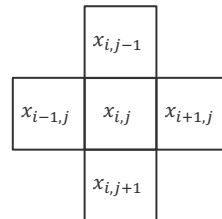
Sampling an Undirected Graphical Model

- **Ising Model:** Defined on an $L \times L$ grid of binary labels $x_{i,j} \in \{-1, +1\}$ with 4-neighbourhood $\mathfrak{N}(i,j) = \{(i+1,j), (i-1,j), (i,j+1), (i,j-1)\}$

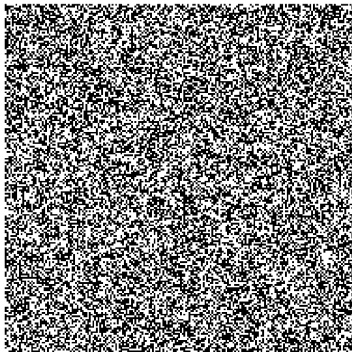
Coupling strength that a pixel has the same label as its neighbours

$$p(\mathbf{x}) \propto \exp \left(\prod_{i=1}^L \prod_{j=1}^L \left[h \cdot x_{i,j} + \sum_{(i',j') \in \mathfrak{N}(i,j)} J \cdot x_{i,j} \cdot x_{i',j'} \right] \right)$$

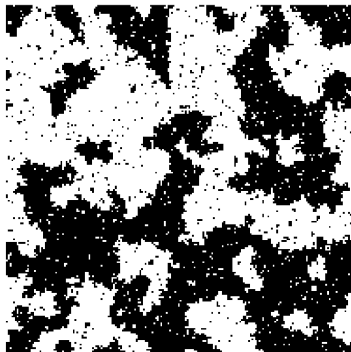
With zero coupling, $h = \frac{1}{2} \cdot \log \left(\frac{p(x_{i,j}=+1)}{1-p(x_{i,j}=+1)} \right)$



$J = 0, h = 0$



$J = 0.5, h = 0$



$J = 1.0, h = 0$



```
# update a single bit using Gibbs sampling
function gibbs_update!(model::IsingModel, i::Int, j::Int)
    L, h, J = model.L, model.h, model.J

    # Calculate the local field
    local_field = h + J * (
        model.x[mod1(i+1, L), j] +
        model.x[mod1(i-1, L), j] +
        model.x[i, mod1(j+1, L)] +
        model.x[i, mod1(j-1, L)]
    )

    # Compute the probability of a state being +1
    p_plus = 1 / (1 + exp(-2 * local_field))

    # Update the single bit
    model.x[i, j] = rand() < p_plus ? 1 : -1

    return model
end
```

■ Graphical Models

- Simple way to visualize the product structure of a joint probability distribution
- Useful for modelling real-life data generating processes
- Allows to test for conditional independence

■ Bayesian Networks

- A directed acyclic graph where each edge points from a conditioning to a conditioned variable in the model (often easier to formulate for experts)
- A generative model of the data that can be easily sampled from (ancestral sampling)
- d-separation is a set of simple rules ("blocking") to read off conditional independence

■ Markov Random Field

- An undirected graph where the probability is proportional to a product of positive potential functions over all cliques
- A generative model of the data that can be easily sampled from (Gibbs sampling)
- Connectivity is the graph property to read off conditional independence

See you next week!